The Birthday Problem

Typesetting Club

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1 The Question

How many people must be in a room before it is more likely than not that two of them share a birthday? Most people guess a number close to half of 365, but the answer is only 23. The question goes back to Mises (1939) and is now a standard example in probability textbooks (Feller 1968). We computed every number in this note with a short script called birthday_table.py.

2 Computing the Probability

Ignore leap years and assume that all 365 birthdays are equally likely. Let p_n be the probability that n people all have different birthdays. The second person must avoid one day, the third must avoid two days, and so on, so

$$p_n = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365 - n + 1}{365} \tag{1}$$

$$= \prod_{k=0}^{n-1} \Big(1 - \frac{k}{365}\Big). \tag{2}$$

Since $1 - x \le e^{-x}$ for every $x \in \mathbb{R}$, equation (1) gives the estimate

$$p_n \le \exp\Big(-\frac{n(n-1)}{730}\Big). \tag{3}$$

The right-hand side falls below $\tfrac{1}{2}$ when $n \ge 23$.

3 Results

Table 1 lists the probability $1 - p_n$ that at least two of n people share a birthday. With 70 people a shared birthday is almost certain.

The result surprises us because we compare ourselves with the others, while the problem counts every pair of people. With 23 people there are already 253 pairs (Diaconis and Mosteller 1989).

Table 1: Probability of a shared birthday

n	$1 - p_n$
10	0.117
20	0.411
23	0.507
30	0.706
50	0.970
70	0.999

References

Diaconis, Persi and Frederick Mosteller (1989). "Methods for Studying Coincidences". In: *Journal of the American Statistical Association* 84.408, pp. 853–861.

Feller, William (1968). *An Introduction to Probability Theory and Its Applications*. 3rd ed. Vol. 1. New York: Wiley.

Mises, Richard von (1939). "Über Aufteilungs- und Besetzungswahrscheinlichkeiten". In: *Revue de la Faculté des Sciences de l'Université d'Istanbul* 4, pp. 145–163.

Verification Log

Bug	**What the AI said**	**Correct?**	**What I did**
B1	The package name is misspelled.	Yes	Wrote amsmath on line 6.
B2			
B3			
B4			
B5			
B6			